Saba Zafar

Curriculum Vitae

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EDUCATION

Master of Science in Computer Science May 2023
University of Sialkot, Sialkot, Pakistan

- Taught in English. Assessed by Germany's Central Office for Foreign Education (ZAB) as equivalent to a German master's degree.
- Relevant coursework: Advanced Machine Learning, Deep Learning, Advanced Human–Computer Interaction, Interaction Design and Analysis, Research Methodology, Natural Language Processing, Information Retrieval.

Bachelor of Education (B.Ed.) 2015
Allama Iqbal Open University, Islamabad, Pakistan — Grade A

Bachelor of Science in Computer Science (Hons.) 2014
National University of Modern Languages, Islamabad, Pakistan

RESEARCH EXPERIENCE

Independent Researcher 2025 – present
Learning analytics and explainable AI in education (unaffiliated)

- Unaffiliated research that produced the two OSF preprints listed below.

Teacher Training Research Scholar (MEXT Scholarship) Oct 2023 – Mar 2025
Graduate School of Education, Aichi University of Education, Kariya, Japan

- Research period Apr 2024 – Mar 2025; major in Educational Administration and Governance; advisor Prof. Tetsuro Ejima.
- Developed the TEACH framework on older teachers' digital teaching skills.

MANUSCRIPTS AND PREPRINTS

Two solo-authored preprints (2026, open access on OSF, not yet peer reviewed):

1. Zafar, S. (2026). *Toward Estimating Cognitive Load and Engagement in Digital Learning Environments Using Behavioral Learning Analytics.* OSF Preprints. https://doi.org/10.35542/osf.io/bgnkw_v1
   - Built random forest and LSTM models on Open University Learning Analytics Dataset (OULAD) clickstream data, then explained their predictions with SHAP.
   - Reports corrected and non-replicating results openly.
2. Zafar, S. (2026). *Explainable and Age-Aware Learning Analytics for Teacher-Facing Systems: A Synthesis and Position Paper.* OSF Preprints. https://doi.org/10.35542/osf.io/25m96_v1

- Argues that teacher-facing analytics should explain their outputs and take into account how teachers of different ages use digital tools.

Unpublished report (2025): the **TEACH framework**, a model of how older in-service teachers acquire digital teaching skills, written during the MEXT research period.

TEACHING AND PROFESSIONAL EXPERIENCE

Secondary School Teacher, Computer Science (full-time) Aug 2017 – Sep 2023
School Education Department, Punjab (Sialkot), Pakistan

- About six years of teaching, with learning-management and interactive tools used in daily lessons.
- Four-week induction training for educators, QAED Sialkot, Aug 2017.

Software Intern Sep 2012 – Feb 2013
NRSoft, Islamabad, Pakistan

- Project planning, feasibility reports, and application development and testing.

GRANTS, FELLOWSHIPS & AWARDS

- Japanese Government (MEXT) Teacher Training Program Scholarship, Oct 2023 – Mar 2025 (individual award)

TECHNICAL SKILLS

Machine learning	Python (PyTorch, scikit-learn); LSTM sequence models, random forests, K-means clustering
Explainability	SHAP explanations of student models
Statistics	Mann-Whitney U, Wilcoxon signed-rank, bootstrap resampling; R (survey package)
Data and visualization	SQL, Tableau, Power BI
Research workflow	LaTeX, Git, Zotero; open science on OSF

LANGUAGES

Urdu (native) · English (professional fluency, C1; all degrees taught in English) · German (intermediate, B1) · Japanese (beginner, A1)

REFERENCES

- **Prof. Tetsuro Ejima**, Education Science Department, ICT Education Infrastructure Center, Aichi University of Education
  tejima@auecc.aichi-edu.ac.jp
- **Prof. Oliver Mayer**, German Language, Aichi University of Education
  omayer@auecc.aichi-edu.ac.jp
- **Ms. Shahnaz Nawaz**, Principal, Government Girls High School Sehjokala, Pakistan
  sehjokalagghs@gmail.com